DS 203 – Big Data Essentials

Advanced SQL Lab 3 – PL/pgSQL

Marks: 100

Student ID: **Student Name:**

Answer the following questions:

Functions and Control Structures

Q1. Write a PL/pgSQL function called get_film_length_category that accepts a film_id and returns a text label based on the length of the film:

- 'Short' if length < 60 minutes
- 'Medium' if length is between 60 and 120 minutes
- 'Long' if length > 120 minutes

Use IF-ELSIF-ELSE control structure.

Code and Usage:

Output Screenshot:

Q2. Write a PL/pgSQL function named `get_customer_status` that accepts a `customer_id` as input and returns the status of the customer. The function should return:

- 'Active' if the customer has rented more than 5 films.
- 'Inactive' if the customer has rented between 1 and 5 films.
- 'New' if the customer has rented no films.

Use the `rental` table to determine the number of films rented by the customer.

Code and Usage:

Output Screenshot:

Functions Only

Q3. Create a PL/pgSQL function `get_customer_full_name` that takes `customer_id` as input and returns the customer's full name (concatenation of `first_name` and `last_name` with a space).

Code and Usage:

Output Screenshot:

Procedures and Control Structures

Q4. Write a stored procedure called `print_long_films` that accepts an integer `min_length` and prints all film titles from the `film` table that have a length greater than or equal to `min_length`. Use a FOR loop and RAISE NOTICE to display each title and the length of each film.

Code and Usage:

Output Screenshot:

Q5. Create a procedure named `show_inventory_by_category` that loops through each film category and prints the category name along with the count of inventory items available in that category (joining `inventory`, `film`, and `category` tables).

Code and Usage:

Output Screenshot:

Marking Criteria:

For each question:

Correct Code and Usage = 15 Marks

Correct Output Screenshot = 5 Marks

Total Marks per Question: 20

Total Marks: 20 x 5 (questions) = 100 Marks.

Submission Instructions:

Submit the following:

1. Convert this document into PDF and rename it as **“DS 203 - Lab 3 – StudentName.pdf”** then submit in the appropriate submission on Google Classroom.
2. Also, submit the .sql script containing codes for all the questions in one single file with comments and explanation guiding the flow of the script. You can name the script file as **“DS 203 - Lab 3 – StudentName.sql”.**